

Random Exercises

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1 Overview

The following document is a compilation of various calculations, computations, and exercises I completed as part of orals prep. The work here is not meant to be an extensive account of my work, by any means.

2 Functional and Harmonic Analysis

2.1 Stein & Shakarchi

Problem 2.2. The following are simple generalizations of the Hausdorff-Young inequalities.

- (a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X, \mu)$. Assume also that $|\varphi_n(x)| \leq M$ for all n . If $a_n = \int f \bar{\varphi}_n d\mu$, then $\|a_n\|_{L^q} \leq M^{(2/p)-1} \|f\|_{L^p(X)}$, $1 \leq p \leq 2$, $1/p + 1/q = 1$.
- (b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \leq \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \leq 1 - 1/p$.

- (a) We initially define the operator T on simple functions f by $Tf = \{a_n\}$, where

$$a_n := \int f \bar{\varphi}_n d\mu.$$

Then clearly, for $(p_0, q_0) = (1, \infty)$, one has that since for each n ,

$$|a_n| \leq \int |f| |\bar{\varphi}_n| d\mu \leq \int M |f| = M \|f\|_{L^1(X)},$$

$\|a_n\|_{L^\infty} = \sup |a_n| \leq M \|f\|_{L^1(X)}$. Thus by Proposition 5.4, we conclude that there exists a unique extension T to all of $L^1(X)$ satisfying the same bound. Likewise, for $(p_1, q_1) = (2, 2)$ and any simple function f , Bessel's inequality implies that

$$\|a_n\|_{L^2} = \left(\sum_{n=1}^{\infty} |a_n|^2 \right)^{1/2} \leq \|f\|_{L^2(X)} = M^{\frac{2}{2}-1} \|f\|_{L^2(X)}.$$

Thus by Proposition 5.4, there exists a unique extension T to all of $L^2(X)$ satisfying the same bound. Then, the Riesz-Thorin Interpolation Theorem tells us that $\|a_n\|_{L^q} \leq M \|f\|_{L^p(X)}$ where

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} = 1 - \frac{1}{2}t, \quad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1} = \frac{1}{2}t,$$

where $t \in [0, 1]$ (thus corresponding to $1 \leq p \leq 2$, $p^{-1} + q^{-1} = 1$), and where

$$M \leq M^{1-t} = M^{2(1-(1/2)t)-1} = M^{\frac{2}{p}-1}.$$

- (b) As before, initially define the operator T on simple functions f on $\mathbb{T}^d$ by $Tf = \{a_n\}$, where for each $n \in \mathbb{Z}^d$,

$$a_n := \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx.$$

Clearly for $(p_0, q_0) = (1, \infty)$, one has

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| = \|f\|_{L^1(\mathbb{T}^d)},$$

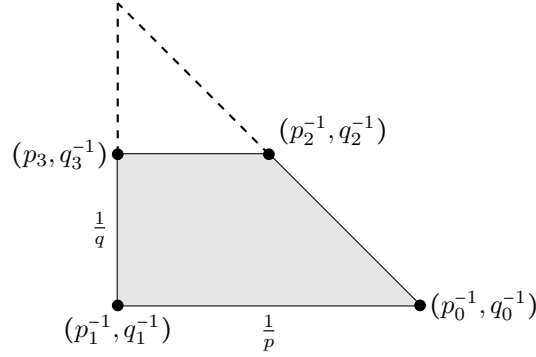
so that

$$\|\{a_n\}\|_{L^\infty(\mathbb{Z}^d)} = \sup_n |a_n| \leq \|f\|_{L^1(\mathbb{T}^d)}.$$

Thus Proposition 5.4 gives a unique extension of T to all $L^1(\mathbb{T}^d)$ that satisfies the same bound. Now consider $(p_1, q_1) = (\infty, \infty)$. Then for any $n \in \mathbb{Z}^d$, one has that

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| \leq \|f\|_{L^\infty(\mathbb{T}^d)},$$

where we assume that dx denotes the normalized measure on $\mathbb{T}^d$. Thus taking the supremum over all $n \in \mathbb{Z}^d$, and using Proposition 5.4 concludes the claim. Likewise, one can show this property holds for the pair $(p_2, q_2) = (2, 2)$. Since $\mathbb{T}^d$ has finite measure, it follows that for all $p \geq 2$ (i.e., $1/p \leq 1/2$), $L^p(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$. Thus, using Proposition 5.4, we also get the endpoint $(p_3, q_3) = (p, 2)$ for all $p \geq 2$. The Riesz diagram as of now is given by



Problem 2.3. Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions f) is possible if and only if $1/p + 1/q = 1$.
 [Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$.]

Let f be an arbitrary fixed simple function. As suggested by the hint, let $r > 0$, and define $f_r(x) = f(rx)$ so that $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$. We note that f_r is also a simple function. We now compute,

$$\begin{aligned} \|\hat{f}_r\|_{L^q(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q r^{-dq} d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\eta)|^q r^d d\eta \right)^{1/q} \\ &= r^{d(-1+1/q)} \|\hat{f}\|_{L^q(\mathbb{R}^d)}. \end{aligned}$$

Likewise,

$$\begin{aligned} \|f_r\|_{L^p(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |f(rx)|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}^d} |f(y)|^p r^{-d} dy \right)^{1/p} = r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)}. \end{aligned}$$

So we have the following: the above estimate holds for all simple functions f if and only if for any $r > 0$,

$$\|\hat{f}_r\|_{L^q(\mathbb{R}^d)} \leq A \|f_r\|_{L^p(\mathbb{R}^d)} \quad \text{and} \quad \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}.$$

But our work above shows that the first inequality is equivalent to the estimate

$$r^{-d+1/q} \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)} \iff \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{d(1-1/q-1/p)} A \|f\|_{L^p(\mathbb{R}^d)}.$$

Thus since $r > 0$ is arbitrary, the above estimate is consistent with the two inequality we have above if and only if $1 - 1/q - 1/p = 0$ iff $1/q + 1/p = 1$.

Problem 2.10. Suppose $f \in L^p(\mathbb{R})$. Verify that:

- (a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \leq \|f\|_{L^p(\mathbb{R})}$, $1 \leq p \leq \infty$.
- (b) $f * \mathcal{P}_y \rightarrow f$, as $y \rightarrow 0$ in the L^p norm when $1 \leq p < \infty$.

(a) Let $\mathcal{P}_y(x)$ denote the **Poisson kernel**,

$$\mathcal{P}_y(x) := \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Let $p_0 = 1$, and define the operator T on $L^{p_0}(\mathbb{R})$, initially just on the set of all simple functions, by $Tf := (f * \mathcal{P}_y)(x)$. First, if $y = 0$, then $\mathcal{P}_y(x) \equiv 0$ so that $Tf \equiv 0$, and the result holds trivially. So assume $y \neq 0$ (wlog, assume $y > 0$). Then we compute,

$$\begin{aligned} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \frac{1}{\pi} \frac{|y|}{x^2 + y^2} dx \\ &= \frac{1}{\pi} \frac{|y|}{y} \arctan\left(\frac{x}{y}\right) \Big|_{-\infty}^{\infty} = \frac{1}{\pi} \frac{|y|}{y} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2} \right] = \frac{|y|}{y} = 1. \end{aligned}$$

Thus for $p_0 = 1$, we have

$$\begin{aligned} \|Tf\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \left| \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x-t)y}{y^2 + t^2} dt \right| dx \\ &\leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dx dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t)\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^1(\mathbb{R})} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^1(\mathbb{R})}, \end{aligned}$$

where the second line follows from Minkowski's inequality, and the fourth line follows from translation invariance of the L^p norms. Thus by Proposition 5.4, T extends to a unique operator on all of $L^1(\mathbb{R})$ satisfying the same estimate. Now define the operator T on $L^{p_1}(\mathbb{R})$, where $p_1 = \infty$, initially on just the set of all simple functions by $Tf := (f * \mathcal{P}_y)(x)$. As before, the proof is trivial if $y = 0$. So assume $y \neq 0$. Then for each x ,

$$\begin{aligned} |Tf(x)| &= \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{y^2 + t^2} dt \right| \\ &\leq \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &\leq \int_{\mathbb{R}} \|f(x-\cdot)\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt = \|f\|_{L^\infty(\mathbb{R})} \cdot \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^\infty(\mathbb{R})}. \end{aligned}$$

Using Proposition 5.4, the result extends to all of $L^\infty(\mathbb{R})$. On the Riesz diagram, we have proven the estimate for the endpoints $(0,0)$ and $(1,1)$. Thus by the Riesz-Thorin interpolation theorem, the theorem follows for all points on the diagonal line connecting these endpoints (i.e., for all $1 \leq p \leq \infty$).

- (b) Now let $1 \leq p < \infty$, and let $\delta > 0$ be fixed. By continuity of translations in $L^p(\mathbb{R})$, there exists $\varepsilon > 0$ such that $|t| \leq \delta \implies \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} < \varepsilon/2$. Thus we compute,

$$\begin{aligned} \|f * \mathcal{P}_y - f\|_{L^p} &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{t^2 + y^2} dt - f(x) \right|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} [f(x-t) - f(x)] \cdot \frac{1}{\pi} \frac{y}{t^2 + y^2} dt \right|^p dx \right)^{1/p} \\ &\leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-t) - f(x)| \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right|^p dx \right)^{1/p} dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &= \int_{|t| \leq \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt + \int_{|t| > \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &\leq \varepsilon/2 + 2 \|f\|_{L^p(\mathbb{R})} \cdot \int_{|t| > \delta} |\mathcal{P}_y(t)| dt, \end{aligned}$$

where the second line follows from $\|\mathcal{P}_y\|_{L^1(\mathbb{R})} = 1$ for any $y \in \mathbb{R}$, and the third line follows from Minkowski's inequality. Since $\mathcal{P}_y(t) \rightarrow 0$ for all $t \neq 0$ in the limit $y \rightarrow 0^+$, it follows that the second term tends to 0 in the limit $y \rightarrow 0^+$. I.e., there exists some y_0 so that for all $y < y_0$, the second term is bounded above by $\varepsilon/(2\|f\|_{L^p(\mathbb{R})})$. Therefore, we conclude that $\|f * \mathcal{P}_{y_0} - f\|_{L^p(\mathbb{R})} \leq \varepsilon$ so that $f * \mathcal{P}_y \rightarrow f$ in the L^p norm.

Problem 2.16. Prove that the space $\mathbf{H}_r^1$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}_r^1$. Then since $\{f_n\}$ is also Cauchy in L^1 , there is an L^1 function f so that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. Now for an appropriate subsequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$.

Let $\mathbf{H}_r^1$ denote the Hardy space, and let $\{f_n\} \subset \mathbf{H}_r^1$ be a Cauchy sequence. First, we begin by showing that each $f \in L^1$. Let $\sum_j \lambda_j \mathbf{a}_j$ be an arbitrary decomposition of f , where each $\mathbf{a}_j$ is an atom, and $\sum |\lambda_j| < \infty$. Thus since the sum converges under the L^1 norm, Fubini/Tonelli implies that

$$\int_{\mathbb{R}} |f| dx = \int_{\mathbb{R}} \left| \sum_1^{\infty} \lambda_k \mathbf{a}_k \right| \leq \int_{\mathbb{R}} \sum_1^{\infty} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} \int_{\mathbb{R}} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} |\lambda_k| < \infty.$$

Thus since $\sum \lambda_j \mathbf{a}_j$ was an arbitrary decomposition of f into a sum of atoms, it follows that $f \in L^1$. In particular, taking the infimum over all such decompositions also yields the estimate $\|f\|_{L^1} \leq \|f\|_{\mathbf{H}_r^1}$. Thus if $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, they are also Cauchy in L^1 . Since the latter is a Banach space, there exists an L^1 function f such that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. But this implies there exists a subsequence $\{f_{n_k}\}$ that converges to f pointwise a.e. Fix $\varepsilon > 0$. Since $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, for every $k \geq 1$, there exists a positive integer N_k such that for all $m, n > N_k$,

$$\|f_m - f_n\|_{\mathbf{H}_r^1} < \frac{\varepsilon}{2^k}.$$

Thus from the subsequence $\{f_{n_k}\}_{k=1}^{\infty}$, we construct another subsequence as follows: we choose $n_1 > N_1$. Next, we choose $n_2 > \max\{n_1, N_2\}$. Repeating inductively, we obtain a subsequence $\{f_{n_k}\}$ such that

$$\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon < \infty.$$

Now we recall that

$$f = \lim_{j \rightarrow \infty} f_{n_j} = f_{n_1} + \lim_{j \rightarrow \infty} \sum_{k=1}^j (f_{n_{k+1}} - f_{n_k}) = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k}).$$

Thus

$$\|f\|_{\mathbf{H}_r^1} \leq \|f_{n_1}\|_{\mathbf{H}_r^1} + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} < \infty,$$

which establishes that $f \in \mathbf{H}_r^1$. All that remains to be shown is that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm. We compute,

$$\|f - f_n\|_{\mathbf{H}_r^1} \leq \|f - f_{n_k}\|_{\mathbf{H}_r^1} + \|f_{n_k} - f_n\|_{\mathbf{H}_r^1}.$$

Since the sequence $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, the second term vanishes in the limit $n, k \rightarrow \infty$. On the other hand, since

$$\|f - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{j=k}^{\infty} \|f_{n_{j+1}} - f_{n_j}\|_{\mathbf{H}_r^1}.$$

By our construction of the subsequence $\{f_{n_k}\}$, we can control the sum by ε . Thus we conclude that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm, which means that $\mathbf{H}_r^1$ is a Banach space.

Problem 2.20. When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \rightarrow \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the “smallness” of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}_r^1$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A \|f\|_{\mathbf{H}_r^1}.$$

[Hint: Verify this for atoms.]

First, assume that for every atom $\mathbf{a}$, the above estimate holds. Then for any decomposition, $f = \sum_{k=1}^{\infty} \lambda_k \mathbf{a}_k$ into atoms,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{\mathbb{R}^d} \left| \sum_1^{\infty} \lambda_k \hat{\mathbf{a}}_k \right| \frac{d\xi}{|\xi|^d} \\ &\leq \sum_1^{\infty} \int_{\mathbb{R}^d} |\lambda_k| |\hat{\mathbf{a}}_k| \frac{d\xi}{|\xi|^d} \\ &\leq A \sum_1^{\infty} |\lambda_k|. \end{aligned}$$

Thus taking the infimum over all such decompositions of f , we obtain,

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \lesssim \|f\|_{\mathbf{H}_r^1}.$$

So now, we will show that the result holds true for every atom. Let $\mathbf{a}$ be an atom on $\mathbb{R}^d$; without loss of generality, we can assume that $\mathbf{a}$ is supported on the unit ball $B_1(0) \subset \mathbb{R}^d$. Assume $\mathbf{a}$ is not identically zero. Then one has that

$$\hat{\mathbf{a}}(\xi) := \int_{\mathbb{R}^d} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) (e^{-2\pi i x \cdot \xi} - 1) dx,$$

where the final equality follows from the fact that $\int_{\mathbb{R}^d} \mathbf{a}(x) dx = 0$. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{|\xi| \leq 1} \left| \int_{|x| \leq 1} \mathbf{a}(x) \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} dx \right| d\xi + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq \int_{|x| \leq 1} \int_{|\xi| \leq 1} |\mathbf{a}(x)| \cdot \left| \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} \right| d\xi dx + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)|^2 d\xi \right)^{1/2} \left(\int_{|\xi| > 1} \frac{1}{|\xi|^{2d}} d\xi \right)^{1/2} \\ &= 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|x| \leq 1} |\mathbf{a}(x)|^2 dx \right)^{1/2} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\frac{1}{m(B_1(0))} \right)^{1/2} \leq 2\pi + \frac{1}{m(B_1(0))^{1/2}} \equiv A < \infty. \end{aligned}$$

2.2 Evans

Problem 2.18. (Stokes' rule) Assume u solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = 0, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v := u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ v = h, v_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is *Stokes' rule*.

Assume u solves the given IVP, and let $v := u_t$. One immediately notes that on $\mathbb{R}^n \times \{t = 0\}$, since $u_t = h$, $v = h$. Second, $v_t = u_{tt}$. On $\mathbb{R}^n \times \{t = 0\}$, $u \equiv 0$ so that $\Delta u = 0$. But since u satisfies the wave equation $u_{tt} = \Delta u \equiv 0$. Thus $v_t \equiv 0$ on $\mathbb{R}^n \times \{t = 0\}$. Next we note that ∂_t and ∂_{x^j} commute for any $1 \leq j \leq n$. Thus since u satisfies $u_{tt} - \Delta u = 0$, differentiating both sides by t , one obtains

$$u_{ttt} - \Delta u_t = 0 \iff v_{tt} - \Delta v = 0$$

in $\mathbb{R}^n \times (0, \infty)$.

Problem 2.19.

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x, y) = F(x) + G(y)$$

for arbitrary functions F, G .

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data g, h is the solution u a right-moving wave? A left-moving wave?

(a) Consider the PDE $u_{xy} = 0$. Let $v(x, y) = u_x$. Then v satisfies the differential equation $v_y = 0$. Thus since v is independent of y , one has $u_x(x, y) = v(x, y) = f(x)$ for some arbitrary function f . But then integrating both sides with respect to x , and supposing that $F(x)$ is the antiderivative of $f(x)$,

$$u(x, y) = F(x) + G(y),$$

where G is some arbitrary function of y .

(b) Define $\xi = x + t$ and $\eta = x - t$ so that $x = (\xi + \eta)/2$ and $t = (\xi - \eta)/2$. Then,

$$\begin{aligned} u_\xi &= u_x \frac{\partial x}{\partial \xi} + u_t \frac{\partial t}{\partial \xi} \\ &= \frac{1}{2} u_x + \frac{1}{2} u_t. \\ u_{\xi\eta} &= \frac{1}{2} \left[u_{xx} \frac{\partial x}{\partial \eta} + u_{xt} \frac{\partial t}{\partial \eta} \right] + \frac{1}{2} \left[u_{tx} \frac{\partial x}{\partial \eta} + u_{tt} \frac{\partial t}{\partial \eta} \right] \\ &= \frac{1}{2} [u_{xx} - u_{xt}] + \frac{1}{2} [u_{tx} - u_{tt}] = \frac{1}{2} [u_{tt} - u_{xx}], \end{aligned}$$

where we used the fact that $u_{xt} = u_{tx}$. Thus, the proof concludes.

(c) Consider the initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Define the change of variables $\xi = x + t$ and $\eta = x - t$ as above. By (b), $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$. By (a), it follows that u must be of the form,

$$u(\xi, \eta) = F(\xi) + G(\eta).$$

Thus, $u(x, t) = F(x + t) + G(x - t)$. Now we must match the initial conditions. At $t = 0$,

$$u(x, 0) = F(x) + G(x) = g(x).$$

$$u_t(x, 0) = F'(x) - G'(x) = h(x).$$

Differentiating the first equation, one first obtains the system

$$\begin{cases} F'(x) + G'(x) = g'(x) \\ F'(x) - G'(x) = h(x). \end{cases}$$

Thus, we get

$$F'(x) = \frac{1}{2}(g'(x) + h(x)) \implies F(x + t) = \frac{1}{2}[g(x + t) - g(0)] + \frac{1}{2} \int_0^{x+t} h(y) dy + F(0).$$

Likewise, we get

$$G(x - t) = \frac{1}{2}[g(x - t) - g(0)] - \int_0^{x-t} h(y) dy + G(0).$$

Adding the two equations, and using the fact that $F(0) + G(0) = g(0)$, we conclude that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy,$$

which is precisely d'Alembert's formula.

- (d) To get a left-moving wave, we want the functions of $(x - t)$ to cancel out. Suppose $h(x) = g'(x)$. Then the fundamental theorem of calculus implies that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} g'(y) dy = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2}[g(x + t) - g(x - t)] = g(x + t).$$

Likewise, to get a right-moving wave, we want the functions of $(x + t)$ to cancel out, so we can take $h(x) = -g'(x)$.

Problem 2.21. (Equipartition of Energy). Let $u \in C^2(\mathbb{R} \times [0, \infty))$ solve the initial-value problem for the wave equation in one-dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Suppose g, h have compact support. The *kinetic energy* is $k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x, t) dx$ and the *potential energy* is $p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x, t) dx$. Prove

- (i) $k(t) + p(t)$ is constant in t ,
(ii) $k(t) = p(t)$ for all large enough times t .

- (i) Let us define the kinetic and potential energy, $k(t)$ and $p(t)$ as given above. We wish to show that the total energy

$$e(t) = k(t) + p(t) = \frac{1}{2} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx$$

is conserved, where u is given by d'Alembert's formula,

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

so that

$$\begin{aligned} u_t(x, t) &= \frac{1}{2}[g'(x+t) - g'(x-t) + h(x+t) + h(x-t)], \\ u_x(x, t) &= \frac{1}{2}[g'(x+t) + g'(x-t) + h(x+t) - h(x-t)]. \end{aligned}$$

Let $T > 0$, and let $K \subset \mathbb{R}$ be a compact set large enough so that $\text{supp } g, \text{supp } h$ are both contained in K . Then it follows that u_t^2 and u_x^2 are both supported in $K_T := K + [-T, T]$ for all $0 \leq t \leq T$. Since u_t^2 and u_x^2 are continuous on $\mathbb{R}$ (and thus on K_T) for all $0 \leq t \leq T$, by the extreme value theorem, there exist finite constants M and N so that for all $x \in K$ and $0 \leq t \leq T$, $|u_t|^2 \leq M\chi_{K_T}$ and $|u_x|^2 \leq N\chi_{K_T}$, where both dominating functions are integrable since K_T is compact. Thus using the Dominated Convergence Theorem, taking the derivative of $e(t)$ gives us,

$$\begin{aligned} \dot{e}(t) &= \frac{1}{2} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx = \frac{1}{2} \frac{\partial}{\partial t} \int_{K_T} u_t^2 + u_x^2 dx \\ &= \frac{1}{2} \int_{K_T} \partial_t(u_t^2 + u_x^2) dx \\ &= \int_{K_T} u_t u_{tt} + u_x \cdot \partial_x u_t dx \\ &= \int_{K_T} u_t(u_{tt} - u_{xx}) dx \quad (\text{by integration by parts}) \\ &= 0, \end{aligned}$$

where there is no boundary term since u_x, u_t are supported inside K_T . So we have shown that $\dot{e}(t) = 0$ for all $0 \leq t \leq T$. Since $T > 0$ was chosen arbitrarily, it follows that $\dot{e}(t) \equiv 0$. Therefore, the total energy is constant in t .

(ii) What we wish is that in the limit $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

First, we compute

$$u_t^2 - u_x^2 = h(x+t)h(x-t) - h(x+t)g'(x-t) + h(x-t)g'(x+t) - g'(x-t)g'(x+t).$$

Then

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_{-\infty}^{\infty} |h(y)h(y-2t)| dy + \int_{-\infty}^{\infty} |h(y)g'(y-2t)| dy \\ &\quad + \int_{-\infty}^{\infty} |h(y-2t)g'(y)| dy + \int_{-\infty}^{\infty} |g'(y-2t)g'(y)| dy. \end{aligned}$$

Let $K \subset \mathbb{R}$ be a compact set large enough that contains the support of both g and h . Then we may write,

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_K |h(y)h(y-2t)| dy + \int_K |h(y)g'(y-2t)| dy \\ &\quad + \int_K |h(y-2t)g'(y)| dy + \int_K |g'(y-2t)g'(y)| dy. \end{aligned}$$

We will only show that the first term on the right vanishes since the work for the remaining terms is identical. Since $h(u) \rightarrow 0$ as $|u| \rightarrow \infty$ (by compact support of h), we know that pointwise $|h(y)h(y-2t)| \rightarrow 0$ as $t \rightarrow \infty$. Next since h is continuous, there exists some finite $M > 0$ so that $|h(y)|, |h(y-2t)| \leq M$ for all $y \in K$. Thus, $|h(y)h(y-2t)| \leq M^2\chi_K \in L^1(\mathbb{R})$ since K is compact. Hence, by the Dominated Convergence Theorem, one has that the first term vanishes as $t \rightarrow \infty$. As we stated before, the remaining three terms also vanish as $t \rightarrow \infty$. Thus, we conclude that as $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$